

Task Tracker patch Notes

① Search filter Bug

- location: `backend/sr/main.java/com/Internet/tasktracker/TaskRepository.java`.
- How I found it: I traced the search request from frontend to repository SQL. The query used AND and OR without parentheses.
- Root cause: SQL evaluates AND before OR. Therefore ~~SQL~~ title matches could bypass the archived & status condition.
- fix: I grouped the title and description conditions:

```
WHERE archived = false
AND status = :status
AND (title LIKE :query OR
description LIKE :query)
```

 This ensure that archived task with wrong status executed correctly.

② pagination validation bug

- location - - - com/Internu/TaskController.java
- How I found it: I reviewed how page and page size were used to calculate like indexes.
- Root cause: Invalid values such as page 0, negative value, very large could create invalid pagination

• fix : I added validation requiring

- $page \geq 1$
- $page size \geq 1$
- $page size \leq 100$

I also made page beyond the ~~available~~ result an empty list safety.

③ Filter and state-request bug

- location: Frontend/src/App.js.
Frontend/src/Hooks/useTask.js

• How I found it: while changing a search or status filter while viewing a latter page. the old page No. remained active.

• Root cause:

A new filter could have fewer pages.
but UI continued requesting previous
page. Also an older slower request
could overwrite the result of
newer request

fix: I reset pagination to page 1
whenever search or status changes.
I also ignore responses from
cancelled or outed requests.

④ Accessibility improvement.

- Location: frontend/src/component/statusfilter.jsx
- Issues: The select did not have
a clear accessible label.
- fix: I added a visible "Status" label
and an aria-label describing
the selector

Verification

- I ran frontend production successfully
- I ran Backend Maven test commands successfully
- I also checked the changed files
for diagnostics and formatting errors.